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HỆ CƠ SỞ DỮ LIỆU Database Systems

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Chapter 4 Entity Relationship (ER) Modeling

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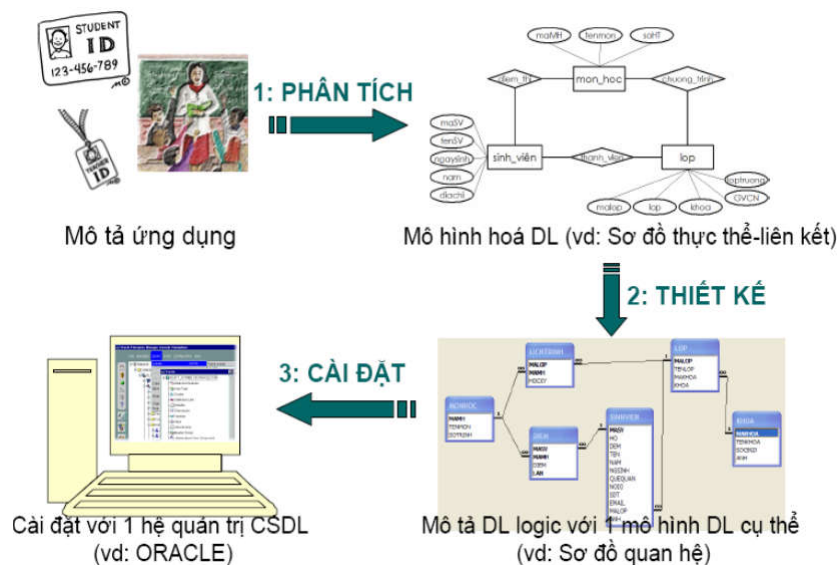
Learning Objectives

- In this chapter, students will learn:
 - The main characteristics of entity relationship components
 - How relationships between entities are defined, refined, and incorporated into the database design process
 - How ERD components affect database design and implementation
 - That real-world database design often requires the reconciliation of conflicting goals

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Các bước xây dựng CSDL



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Entity Relationship Model (ERM)

- Basis of an entity relationship diagram (ERD)
- ERD depicts the:
 - Conceptual database as viewed by end user
 - Database's main components
 - Entities
 - Attributes
 - Relationships

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Entities

- An entity is an object of interest to the end user
- Entity - Refers to the entity set and not to a single entity occurrence
- In other words, an entity in the ERM corresponds to a table—not to a row—in the relational environment.
- The ERM refers to a table row as an entity instance or entity occurrence.

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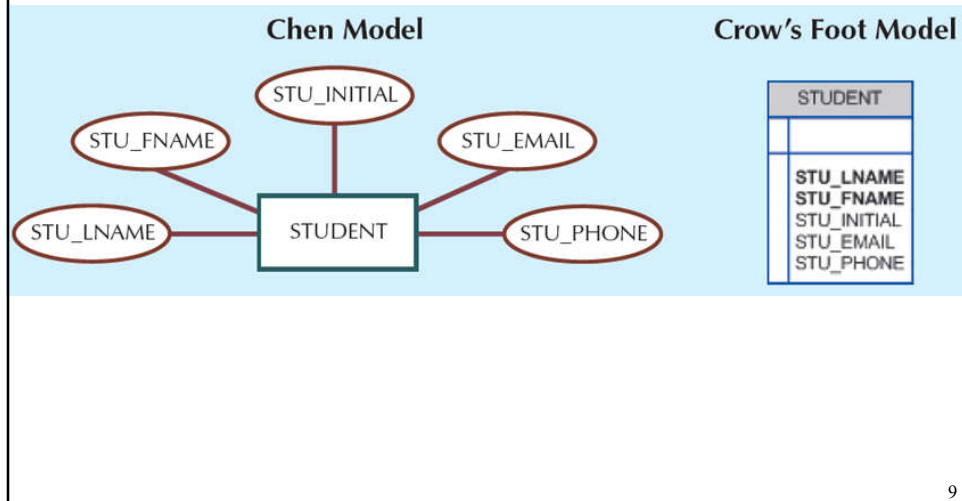
Entities

- In the Chen, Crow's Foot, and UML notations, an entity is represented by a rectangle that contains the entity's name.
- The entity name, a noun, is usually written in all capital letters.

Attributes

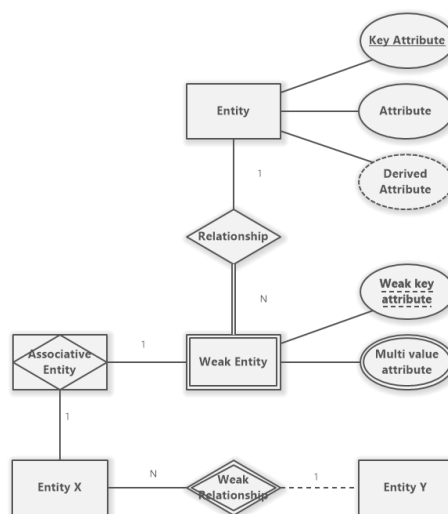
- Characteristics of entities
- In the original Chen notation, attributes are represented by ovals and are connected to the entity rectangle with a line. Each oval contains the name of the attribute it represents.
- In the Crow's Foot notation, the attributes are written in the attribute box below the entity rectangle.

Figure 4.1 - The Attributes of the Student Entity:
Chen and Crow's Foot

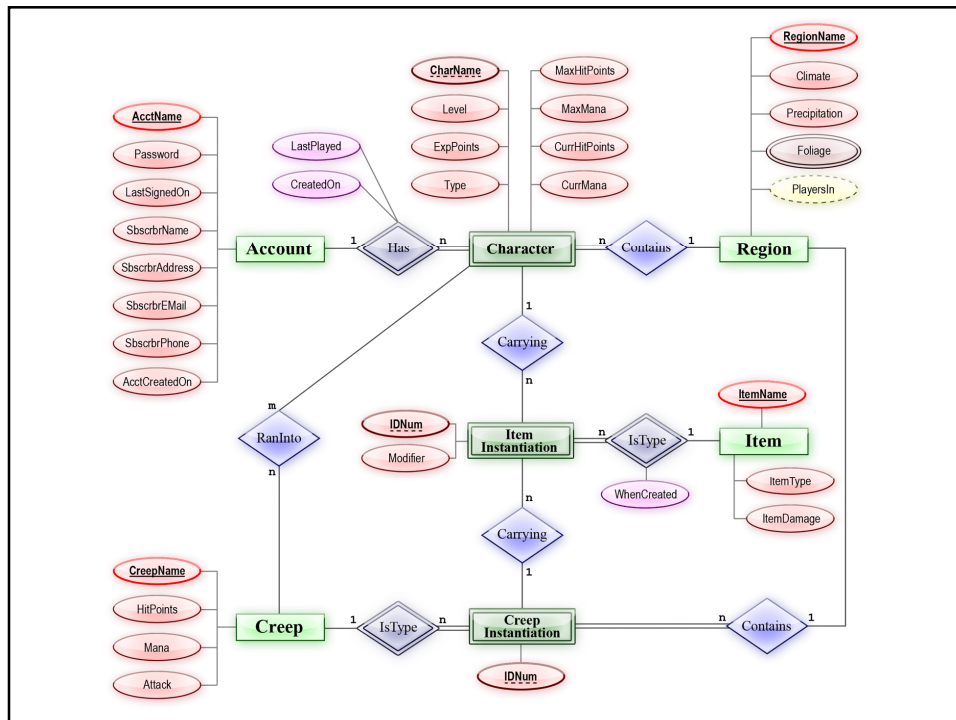


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ER example

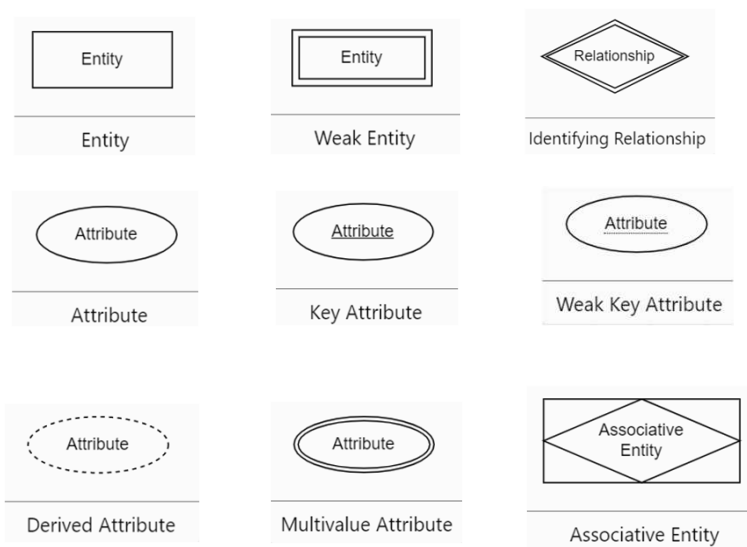


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Chen Notation Basic



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Required and Optional Attributes

- **Required attribute:** Must have a value, cannot be left empty
 - STU_LNAME and STU_FNAME require data entries
- **Optional attribute:** Does not require a value, can be left empty
 - Students might not have a middle name, and perhaps they do not yet have a phone number and an email address.

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Domains

- Attributes have a domain.
- A domain is the set of possible values for a given attribute.
 - For example, the domain for a grade point average (GPA) attribute is written (0,4)
 - The domain for a gender attribute consists of only two possibilities: M or F (or some other equivalent code)...

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Identifiers (Primary Keys)

- **Identifiers—one or more attributes that uniquely identify each entity instance.**
- In the relational model, entities are mapped to tables, and the entity identifier is mapped as the table's primary key (PK).
- Identifiers are underlined in the ERD.
- Key attributes are also underlined in a frequently used shorthand notation for the table structure, called a relational schema

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Relational schema

- TABLE NAME (KEY ATTRIBUTE 1, ATTRIBUTE 2, ATTRIBUTE 3, ... ATTRIBUTE K)
- **For example**
- CAR (CAR VIN, MOD_CODE, CAR_YEAR, CAR_COLOR)
- Each car is identified by a unique vehicle identification number, or CAR_VIN

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Composite Identifiers

- A primary key composed of more than one attribute
- For instance:
 - the Tiny College database administrator may decide to identify each CLASS entity instance (occurrence)
 - by using a composite primary key of **CRS_CODE** and **CLASS_SECTION**
 - instead of using CLASS_CODE

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Composite Identifiers

FIGURE 4.2 THE CLASS TABLE (ENTITY) COMPONENTS AND CONTENTS

Database name: Ch04_TinyCollege

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10012	ACCT-211	1	MMWF 8:00-8:50 a.m.	BUS311	105
10013	ACCT-211	2	MMWF 9:00-9:50 a.m.	BUS200	105
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10015	ACCT-212	1	MMWF 10:00-10:50 a.m.	BUS311	301
10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
10017	CIS-220	1	MMWF 9:00-9:50 a.m.	KLR209	228
10018	CIS-220	2	MMWF 9:00-9:50 a.m.	KLR211	114
10019	CIS-220	3	MMWF 10:00-10:50 a.m.	KLR209	228
10020	CIS-420	1	vv 6:00-8:40 p.m.	KLR209	162
10021	QM-261	1	MMWF 8:00-8:50 a.m.	KLR200	114
10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
10023	QM-362	1	MMWF 11:00-11:50 a.m.	KLR200	162
10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162
10025	MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325

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Composite Identifiers

- If the CLASS_CODE in Figure 4.2 is used as the primary key, the CLASS entity may be represented in shorthand form as follows:
- CLASS (CLASS_CODE, CRS_CODE, CLASS_SECTION, CLASS_TIME, ROOM_CODE, PROF_NUM)
- On the other hand, if CLASS_CODE is deleted, and the composite primary key is the combination of CRS_CODE and CLASS_SECTION, the CLASS entity may be represented as follows:
- CLASS (CRS_CODE, CLASS_SECTION, CLASS_TIME, ROOM_CODE, PROF_NUM)

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Attributes

- **Composite attribute:** Attribute that can be subdivided to yield additional attributes
- **Simple attribute:** Attribute that cannot be subdivided
- **Single-valued attribute:** Attribute that has only a single value
- **Multivalued attributes:** Attributes that have many values

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Composite and Simple Attributes

- A **composite attribute**, not to be confused with a composite key, is an attribute that can be further subdivided to yield additional attributes.
 - For example:
 - The attribute ADDRESS can be subdivided into street, city, state, and zip code.
 - The attribute PHONE_NUMBER can be subdivided into area code and exchange number.
- A **simple attribute** is an attribute that cannot be subdivided.
 - For example, age, sex, and marital status would be classified as simple attributes.

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Single-Valued Attributes

- A **single-valued attribute is an attribute that can have only a single value.**
 - For example:
 - a person can have only one Social Security number,
 - a manufactured part can have only one serial number.
 - *Keep in mind that a single-valued attribute is not necessarily a simple attribute.* For instance, a part's serial number (such as SE-08-02-189935) is single-valued, but it is a composite attribute...

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Multivalued Attributes

- Multivalued attributes are attributes that can have many values.
- For instance
 - a person may have several college degrees,
 - a household may have several different phones, each with its own number
 - a car's color may be subdivided into many colors for the roof, body, and trim

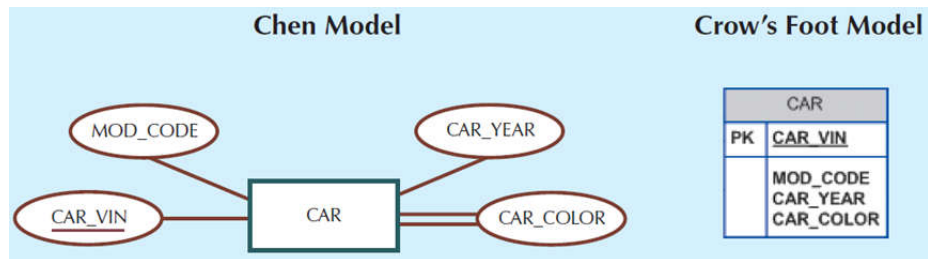
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Multivalued Attributes

- In the Chen ERM, multivalued attributes are shown by a double line connecting the attribute to the entity.
- The Crow's Foot notation does not identify multivalued attributes.

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Figure 4.3 - A Multivalued Attribute in an Entity

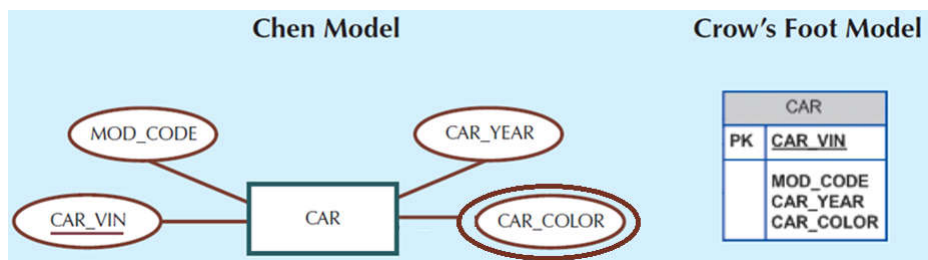


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Implementing Multivalued Attributes

- Although the conceptual model can handle M:N relationships and multivalued attributes, you should not implement them in the RDBMS.
- Remember from Chapter 3 that in the relational table, each column and row intersection represents a single data value

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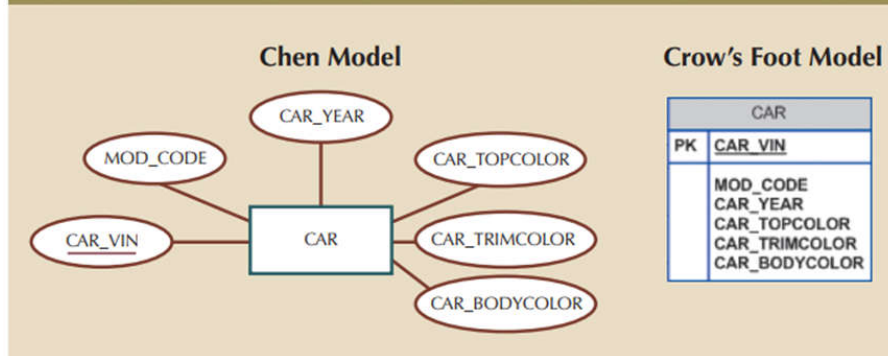
Implementing Multivalued Attributes

- If multivalued attributes exist, the designer must decide on one of two possible courses of action:
 - ❶ **Within the original entity, create several new attributes, one for each component of the original multivalued attribute.**
 - For example, the CAR entity's attribute CAR_COLOR can be split to create the new attributes CAR_TOPCOLOR, CAR_BODYCOLOR, and CAR_TRIMCOLOR, which are then assigned to the CAR entity. (See Figure 4.4.)

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Implementing Multivalued Attributes

FIGURE 4.4 SPLITTING THE MULTIVALUED ATTRIBUTE INTO NEW ATTRIBUTES



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Implementing Multivalued Attributes

- Although this solution seems to work, its adoption can lead to major structural problems in the table. It is only acceptable if every instance will have the same number of values for the multivalued attribute, and no instance will ever have more values.
- If some employees have 10 degrees and certifications while most have fewer or none, the number of degree/certification attributes would be 10, and most of those attribute values would be null for most employees). In short, although you have seen solution 1 applied, it is not always acceptable.

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Implementing Multivalued Attributes

② Create a new entity composed of the original multivalued attribute's components.

This new entity allows the designer to define color for different sections of the car (see Table 4.1). Then, this new CAR_COLOR entity is related to the original CAR entity in a 1:M relationship.

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Implementing Multivalued Attributes

TABLE 4.1

COMPONENTS OF THE MULTIVALUED ATTRIBUTE

SECTION	COLOR
Top	White
Body	Blue
Trim	Gold
Interior	Blue

FIGURE 4.5 A NEW ENTITY SET COMPOSED OF A MULTIVALUED ATTRIBUTE'S COMPONENTS



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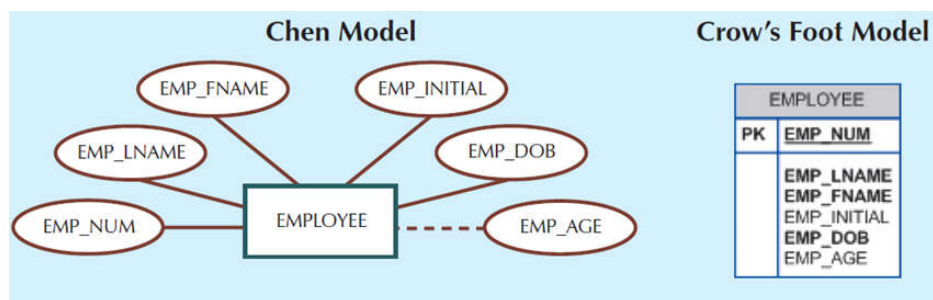
Derived attribute

- Attribute whose value is calculated from other attributes
 - Derived using an algorithm
 - Derived attributes are sometimes referred to as *computed* attributes

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Figure 4.6 - Depiction of a Derived Attribute

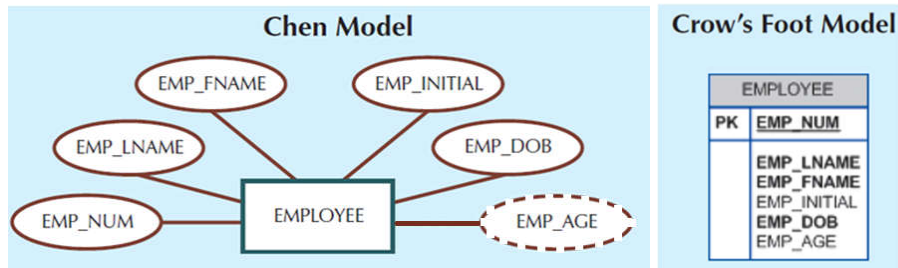


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Table 4.2 - Advantages and Disadvantages of Storing Derived Attributes

	Stored	Not stored
Advantage	- Saves CPU processing cycles - Saves data access time - Data value is readily available - Can be used to keep track of historical data	- Saves storage space - Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	- Uses CPU processing cycles - Increases data access time - Adds coding complexity to queries

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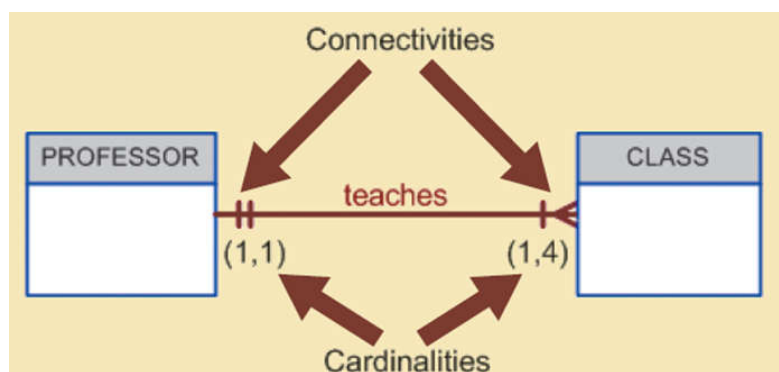
Relationships

- Association between entities that always operate in both directions
- **Participants:** Entities that participate in a relationship
- **Connectivity:** Describes the relationship classification
- **Cardinality:** Expresses the minimum and maximum number of entity occurrences associated with one occurrence of related entity

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Figure 4.7 - Connectivity and Cardinality in an ERD



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Existence Dependence

Existence dependence

- Entity exists in the database only when it is associated with another related entity occurrence

Existence independence

- Entity exists apart from all of its related entities
- Referred to as a **strong entity** or **regular entity**

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Relationship Strength

Weak (non-identifying) relationship

- Primary key of the related entity does not contain a primary key component of the parent entity

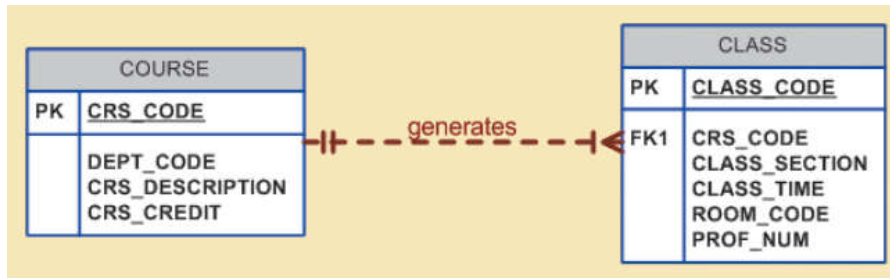
Strong (identifying) relationships

- Primary key of the related entity contains a primary key component of the parent entity

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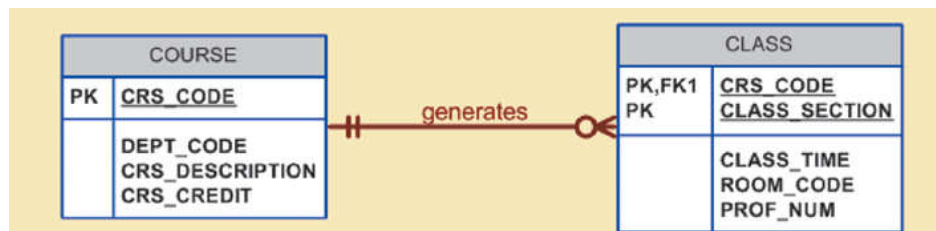
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Figure 4.8 - A Weak (Non-Identifying) Relationship between COURSE and CLASS



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Figure 4.9 - A Strong (Identifying) Relationship between COURSE and CLASS



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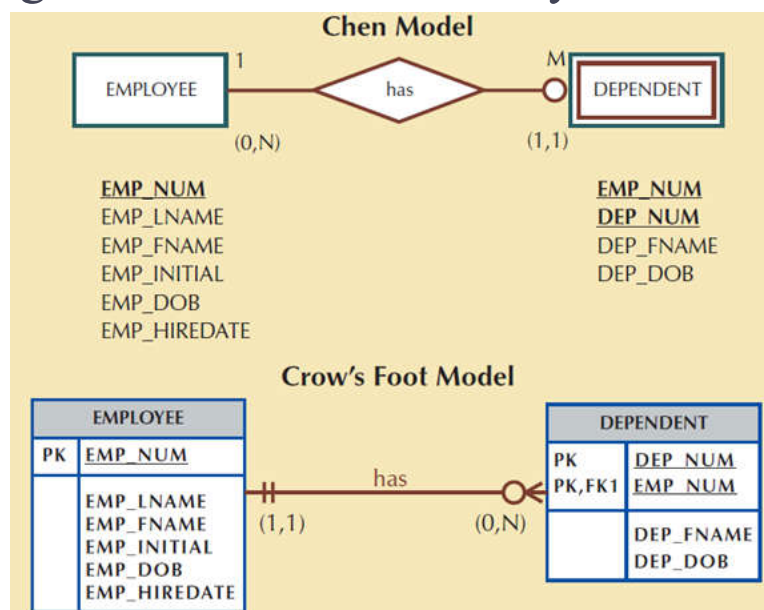
Weak Entity

- Conditions
 - Existence-dependent
 - Has a primary key that is partially or totally derived from parent entity in the relationship
- Database designer determines whether an entity is weak based on business rules

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Figure 4.10 - A Weak Entity in an ERD



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Figure 4.11 - A Weak Entity in a Strong Relationship

Table name: EMPLOYEE				Database name: Ch04_ShortCo	
EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_HIREDATE
1001	Callifante	Jeanine	J	12-Mar-64	25-May-97
1002	Smithson	William	K	23-Nov-70	28-May-97
1003	Washington	Herman	H	15-Aug-68	28-May-97
1004	Chen	Lydia	B	23-Mar-74	15-Oct-98
1005	Johnson	Melanie		28-Sep-66	20-Dec-98
1006	Ortega	Jorge	G	12-Jul-79	05-Jan-02
1007	O'Donnell	Peter	D	10-Jun-71	23-Jun-02
1008	Brzenski	Barbara	A	12-Feb-70	01-Nov-03

Table name: DEPENDENT			
EMP_NUM	DEP_NUM	DEP_FNAME	DEP_DOB
1001	1	Annelise	05-Dec-97
1001	2	Jorge	30-Sep-02
1003	1	Suzanne	25-Jan-04
1006	1	Carlos	25-May-01
1008	1	Michael	19-Feb-95
1008	2	George	27-Jun-98
1008	3	Katherine	18-Aug-03

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Relationship Participation

Optional participation

- One entity occurrence does not require a corresponding entity occurrence in a particular relationship

Mandatory participation

- One entity occurrence requires a corresponding entity occurrence in a particular relationship

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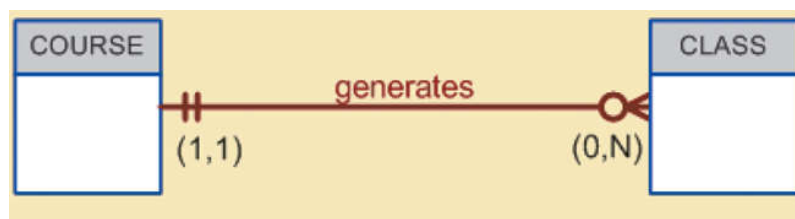
Table 4.3 - Crow's Foot Symbols

CROW'S FOOT SYMBOLS	CARDINALITY	COMMENT
	(0,N)	Zero or many; the "many" side is optional.
	(1,N)	One or many; the "many" side is mandatory.
	(1,1)	One and only one; the "1" side is mandatory.
	(0,1)	Zero or one; the "1" side is optional.

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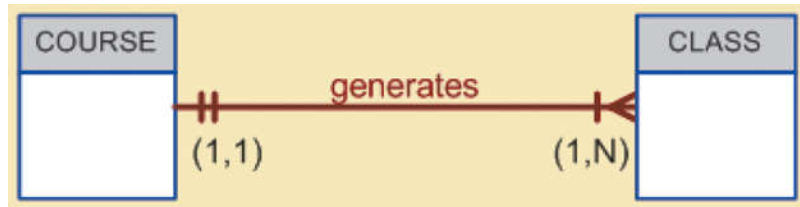
Figure 4.13 - CLASS is **Optional** to COURSE



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Figure 4.14 - COURSE and CLASS in a
Mandatory Relationship



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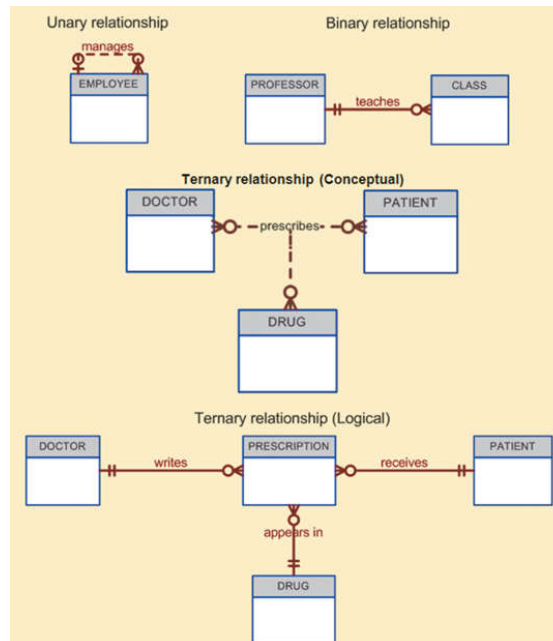
Relationship Degree

- Indicates the number of entities or participants associated with a relationship
- **Unary relationship:** Association is maintained within a single entity
 - **Recursive relationship:** Relationship exists between occurrences of the same entity set
- **Binary relationship:** Two entities are associated
- **Ternary relationship:** Three entities are associated

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Figure 4.15 - Three Types of Relationship Degree



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Figure 4.17 - An ER Representation of Recursive Relationships



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Associative Entities

- Also known as composite or bridge entities
- Used to represent an M:N relationship between two or more entities
- Is in a 1:M relationship with the parent entities
 - Composed of the primary key attributes of each parent entity
- May also contain additional attributes that play no role in connective process

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Figure 4.23 - Converting the M:N Relationship into Two 1:M Relationships

Table name: STUDENT		Database name: Ch04_CollegeTry			
STU_NUM	STU_LNAME				
321452	Bowser				
324257	Smithson				

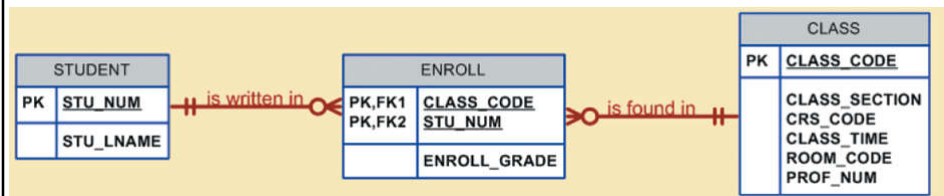
Table name: ENROLL					
CLASS_CODE	STU_NUM	ENROLL_GRADE			
10014	321452	C			
10014	324257	B			
10018	321452	A			
10018	324257	B			
10021	321452	C			
10021	324257	C			

Table name: CLASS					
CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MWTF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MWTF 8:00-8:50 a.m.	KLR200	114

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Figure 4.25 - A Composite Entity in an ERD



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Developing an ER Diagram

- The process of database design is iterative rather than a linear or sequential process.
- The verb *iterate* means "to do again or repeatedly." Thus, an **iterative process** is based on repetition of processes and procedures.
- Building an ERD usually involves the following activities:

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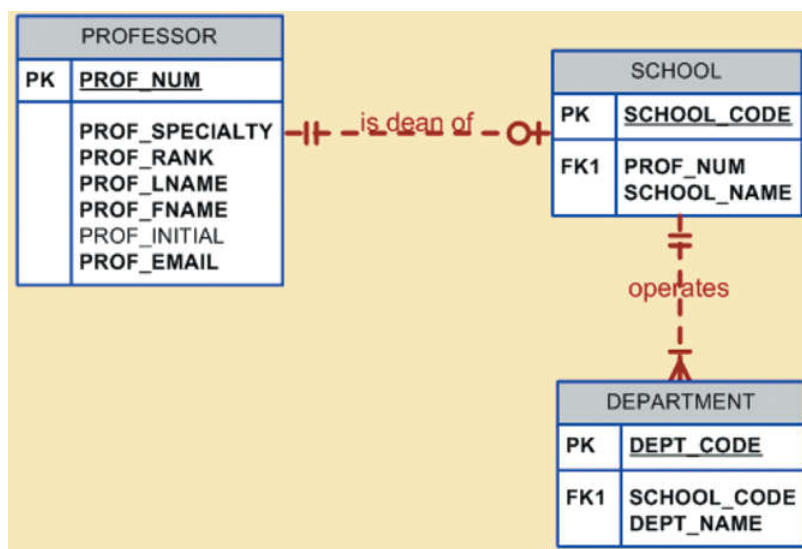
Developing an ER Diagram

- Create a detailed narrative of the organization's description of operations
- Identify business rules based on the descriptions
- Identify main entities and relationships from the business rules
- Develop the initial ERD
- Identify the attributes and primary keys that adequately describe entities
- Revise and review ERD

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Figure 4.26 - The First Tiny College ERD Segment



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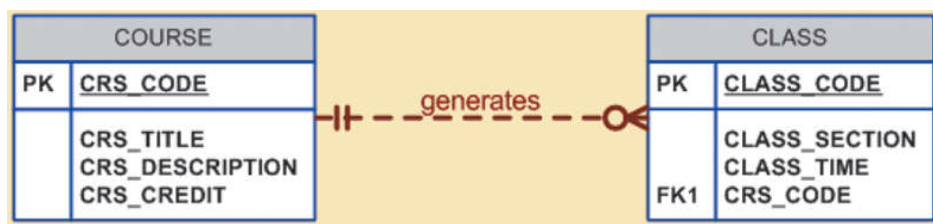
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Figure 4.27 - The Second Tiny College ERD Segment



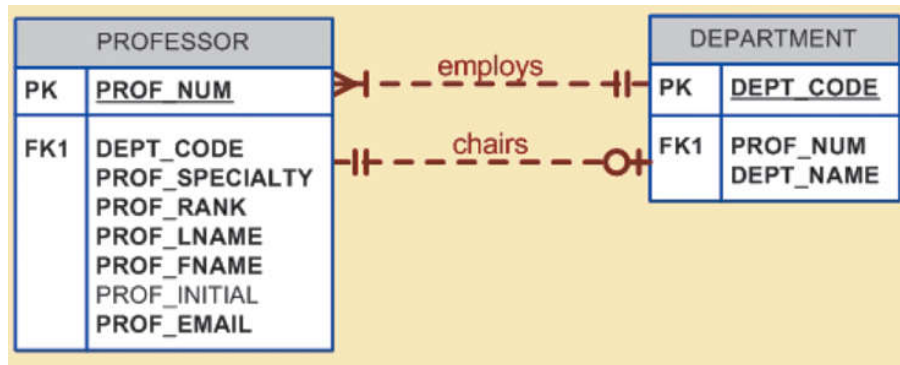
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Figure 4.28 - The Third Tiny College ERD Segment



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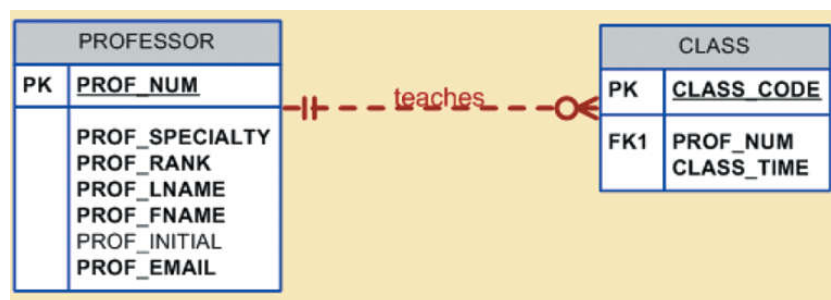
Figure 4.29 - The Fourth Tiny College ERD Segment



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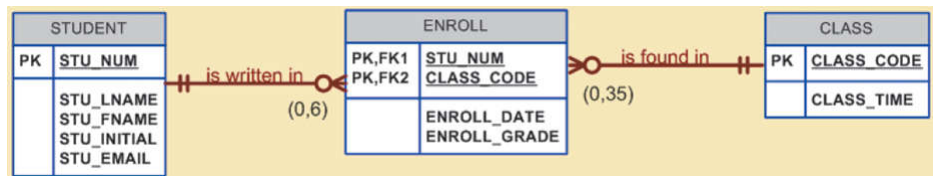
Figure 4.30 - The Fifth Tiny College ERD Segment



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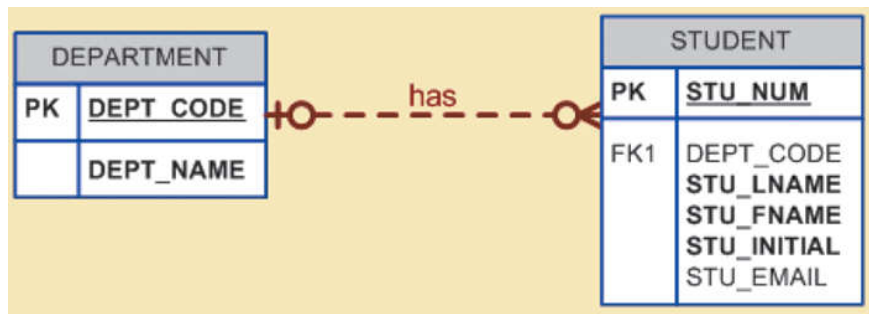
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Figure 4.31 - The Sixth Tiny College ERD Segment



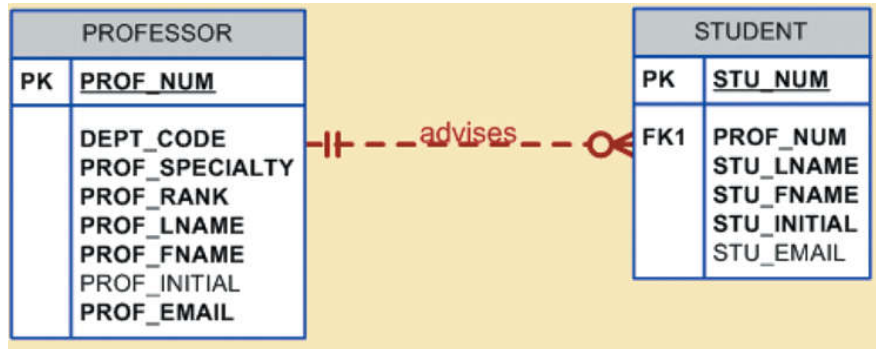
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Figure 4.32 - The Seventh Tiny College ERD Segment



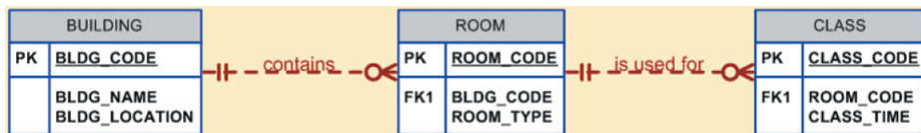
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Figure 4.33 - The Eighth Tiny College ERD Segment



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Figure 4.34 - The Ninth Tiny College ERD Segment



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Table 4.4 - Components of the ERM

ENTITY	RELATIONSHIP	CONNECTIVITY	ENTITY
SCHOOL	operates	1:M	DEPARTMENT
DEPARTMENT	has	1:M	STUDENT
DEPARTMENT	employs	1:M	PROFESSOR
DEPARTMENT	offers	1:M	COURSE
COURSE	generates	1:M	CLASS
PROFESSOR	is dean of	1:1	SCHOOL
PROFESSOR	chairs	1:1	DEPARTMENT
PROFESSOR	teaches	1:M	CLASS
PROFESSOR	advises	1:M	STUDENT
STUDENT	enrolls in	M:N	CLASS
BUILDING	contains	1:M	ROOM
ROOM	is used for	1:M	CLASS

Note: ENROLL is the composite entity that implements the M:N relationship "STUDENT enrolls in CLASS."

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Database Design Challenges: Conflicting Goals

Database design must conform to design standards

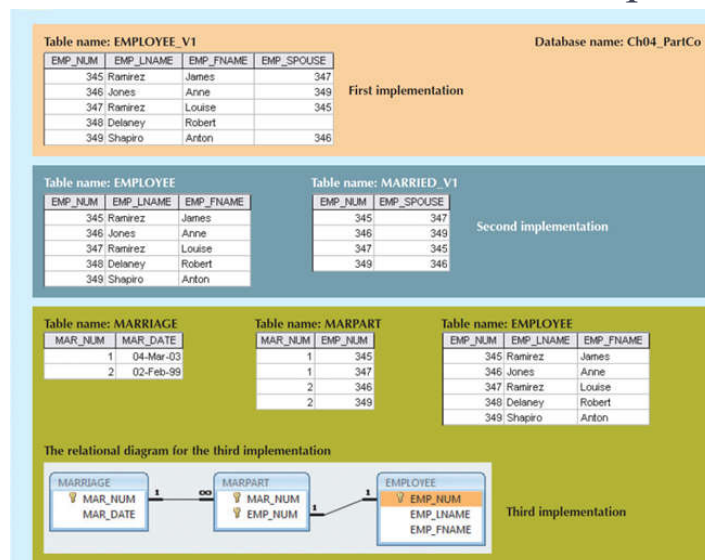
Need for high processing speed may limit the number and complexity of logically desirable relationships

Need for maximum information generation may lead to loss of clean design structures and high transaction speed

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Figure 4.38 - Various Implementations of the 1:1 Recursive Relationship



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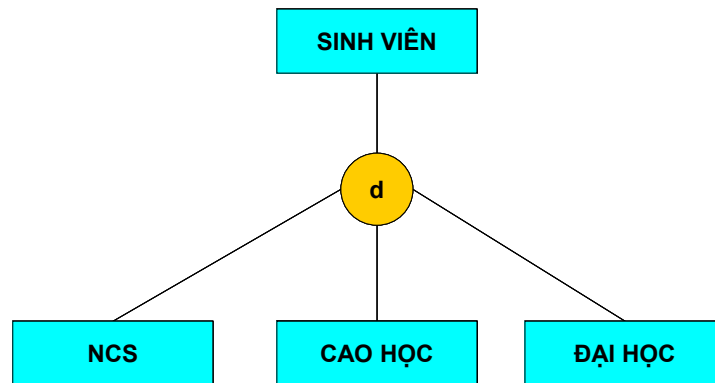
Phân cấp là một (ISA)

Phân cấp "là một" : Cho phép thể hiện sự kế thừa giữa các tập thực thể

Tập thực thể A "là một" tập thực thể con (sub-entity) của Tập thực thể B nếu:

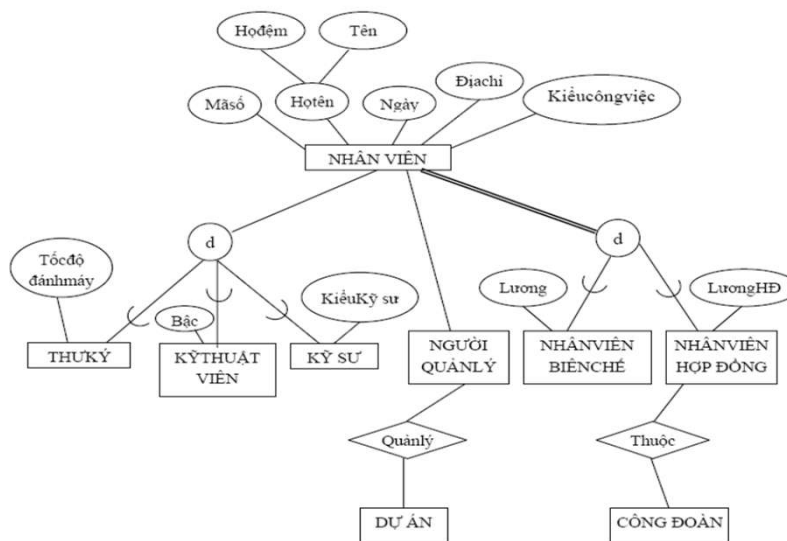
- A kế thừa đầy đủ các thuộc tính của B
- A có một vài thuộc tính riêng
- Có tất cả các liên kết liên quan đến B
- Có một vài liên kết riêng

Phân cấp là một (ISA)



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Phân cấp là một (ISA)



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